Project Design Phase

Solution Architecture

Date	21 October 2026
Team ID	Individual Project
Project Name	StaffSync – Modern Human Resource Management System
Maximum Marks	4 Marks

Solution Architecture

StaffSync is a web-based Human Resource Management System (HRMS) developed using the MERN stack. Its architecture separates the user interface, application logic, security controls, and data storage. Employees, Managers, and HR users access role-appropriate features through a React frontend. The frontend communicates with a Node.js and Express backend through REST APIs, while MongoDB stores the application data through Mongoose models.

Main Architectural Components

- **Presentation layer:** React-based interface with React Router for navigation, Redux Toolkit for shared client state, and Axios for API requests.
- **Application layer:** Node.js and Express provide REST endpoints and business logic for employee profiles, attendance, leave requests, announcements, training, and performance workflows.
- **Authentication and authorization:** JWT-based authentication, password hashing with bcryptjs, protected routes, input validation, and role-based access for Employee, Manager, and HR users.
- **Data layer:** MongoDB with Mongoose models for users, employees, managers, attendance, leave, announcements, training, notifications, and performance.
- **Configuration and deployment:** Environment variables hold sensitive configuration such as database connection strings and JWT secrets. Deployment depends on the configured hosting environment; no specific production hosting platform is assumed here.

Typical Request Flow

1. A user signs in and receives an authentication token after the credentials are validated.
2. The React interface sends requests to the Express REST API using Axios.
3. The backend validates the request, verifies authentication and role permissions, and applies the relevant business rules.
4. Mongoose reads or updates the relevant MongoDB collections.
5. The API returns a response to the frontend, which updates the displayed page or status.

Architecture Diagram

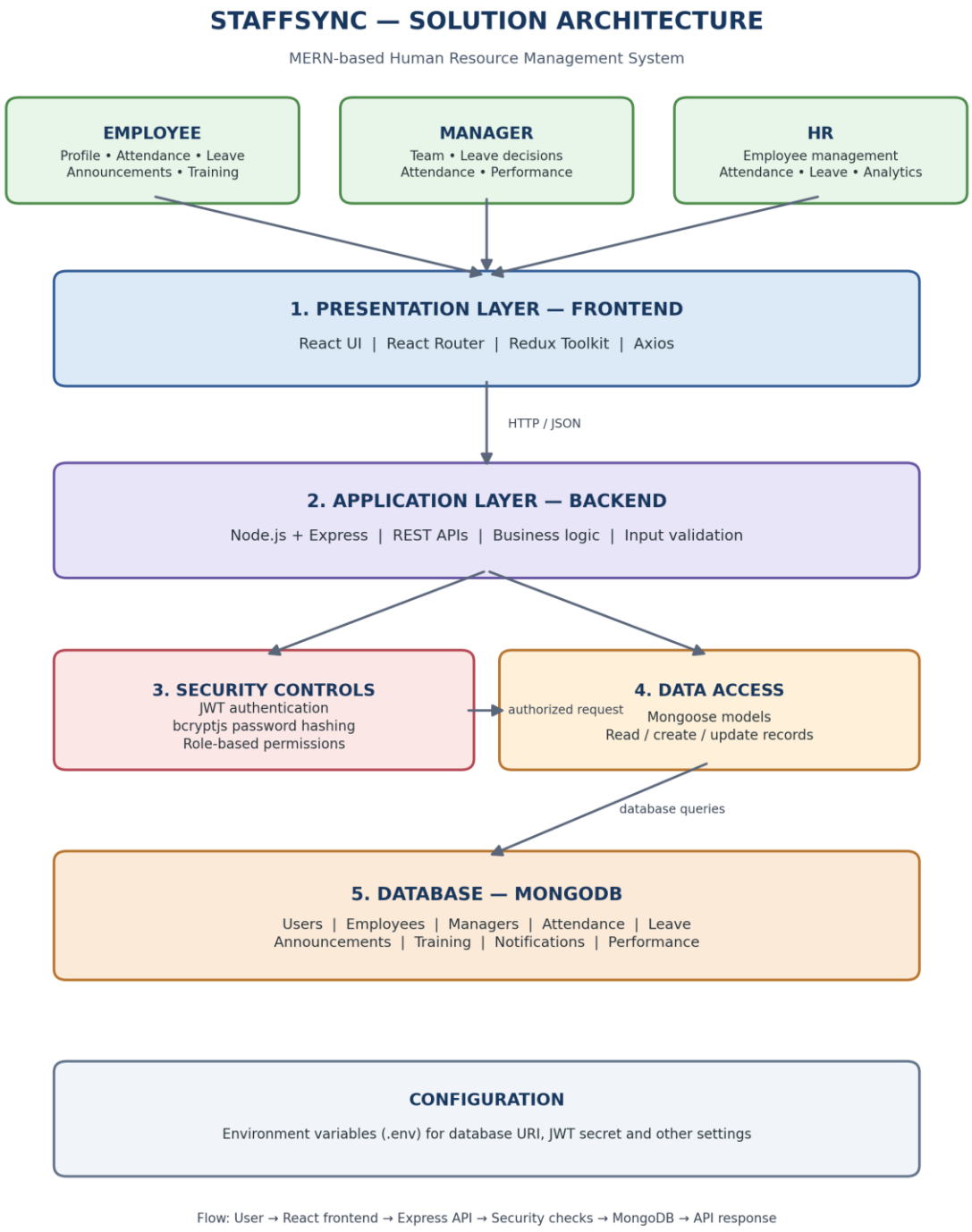


Figure: High-level architecture of StaffSync

Key Design Considerations

- Separation of frontend, backend, and database responsibilities makes the system easier to maintain.
- Role-based authorization restricts employee, manager, and HR operations to the permissions assigned to each role.
- Server-side validation and secure password hashing help protect account and workflow data.
- The data model supports HR workflows including attendance, leave management, announcements, training, notifications, and performance tracking.

- Features such as payroll processing, biometric attendance, AI-based recommendations, and video calls are not represented as implemented features in this architecture.